



Smart Emergency Shelter & Safe Route Navigation System

Presented by: Team ICONIX

INTRODUCE TEAM MEMBERS

Mg Htet Oo Wai Yan
Mg Saw Yan Naing Moe
Mg Hein Yar Zar
Ma Thiri Min Zaw
Ma Myint Myat Kyi
Ma Myat Noe Shwe Sin
Ma Moe Myat Myat Khaing



Problems During Emergencies

- Difficult to find the nearest shelter or hospital
- Unclear where to go in unfamiliar areas
- Time wasted searching for help
- Risk of choosing unsafe or longer routes
- Emergency locations are not centralized.

Why This Problem Matters

During a disaster, people need clear information and fast decisions, not just a map. Even when shelters and hospitals are available, people may not know:

- which one is closest,
- how far it is,
- or how to reach it.

SafePath was designed to connect people, safe places, and emergency support in one system.

SafePath Solution

SafePath is a location-based emergency navigation system that:

- Detects the user's current location.
- Shows nearby shelters and hospitals on the map.
- Calculates the distance to emergency locations.
- Finds the nearest available safe place.
- Provides directions to reach the selected location.
- Allows users to request help through the “I Need Help” function.

How SafePath Works

Current Location → Nearby Safe Places →
Distance Calculation → Best Location →
Route & Direction

When the user presses “I Need Help”, the system uses their current location to identify the nearest suitable shelter or hospital and provides a route to reach it.

Smart Features

-  Real-Time User Location
-  Interactive Location Map
-  Shelter & Hospital Markers
-  Distance Calculation
-  “I Need Help” Emergency Button
-  Nearest Safe Place Recommendation
-  Route & Direction Guidance
-  Emergency Camp Management
-  Admin Dashboard

Benefits of SafePath

SafePath helps users:

- Find safe locations faster.
- Reduce confusion during emergencies.
- Avoid unnecessary travel.
- Reduce the time needed to search for help.
- Reach shelters or hospitals using clear directions.
- Make faster and more informed decisions.

For administrators, the system provides a centralized way to manage and update emergency camps.

Smart Technology in SafePath

SafePath combines:

Location Services + Digital Mapping + Distance
Calculation + Route Recommendation + Emergency Data
Management

Instead of only displaying emergency locations, the system uses the user's position to calculate which location is the most practical to reach

Supporting a Smarter & Greener Community

SafePath supports sustainable emergency management by:

- Reducing unnecessary travel during disasters.
- Helping people use nearby existing shelters and hospitals efficiently.
- Improving the use of emergency resources.
- Supporting organized and location-based disaster response.
- Reducing inefficient movement and congestion around emergency areas.

Expected Impact

For Citizens:

Faster access to safety and emergency support.

For Emergency Teams:

Better organization of shelters and camps.

For Communities:

More efficient use of available resources.

Overall:

Less Confusion • Faster Response • Smarter Navigation
• Safer Communities

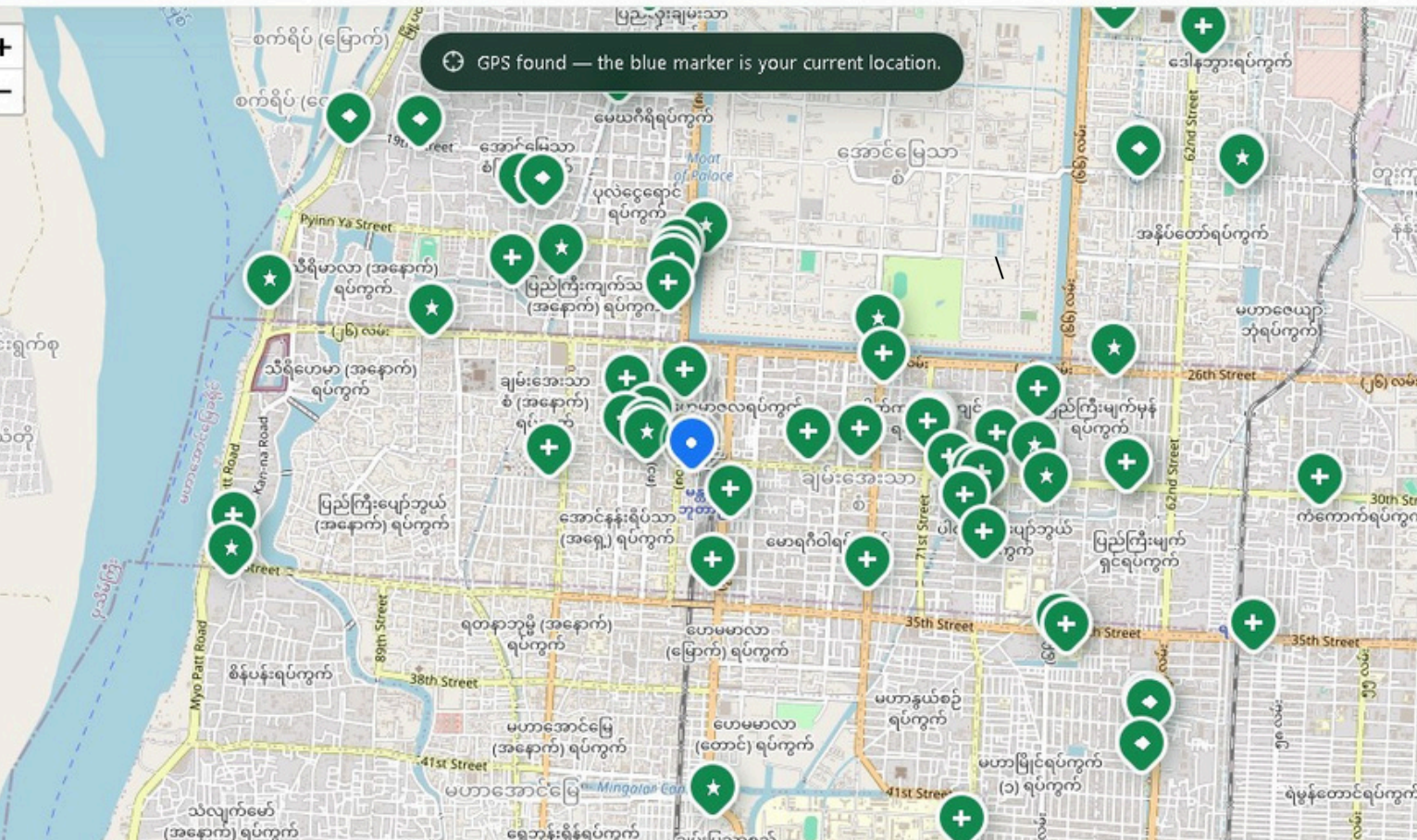
HOME PAGE

→ ↺ ⓘ localhost:3000

 **SafePath**
REAL-TIME EMERGENCY RESPONSE

📍 EN Admin

📍 GPS found — the blue marker is your current location.



LOCATION DETAILS

Find the safest next step.

Allow GPS to see your blue marker and real nearby emergency locations.



Tap any green or orange marker to see its name, distance, status, resources, and directions.

Nearest locations

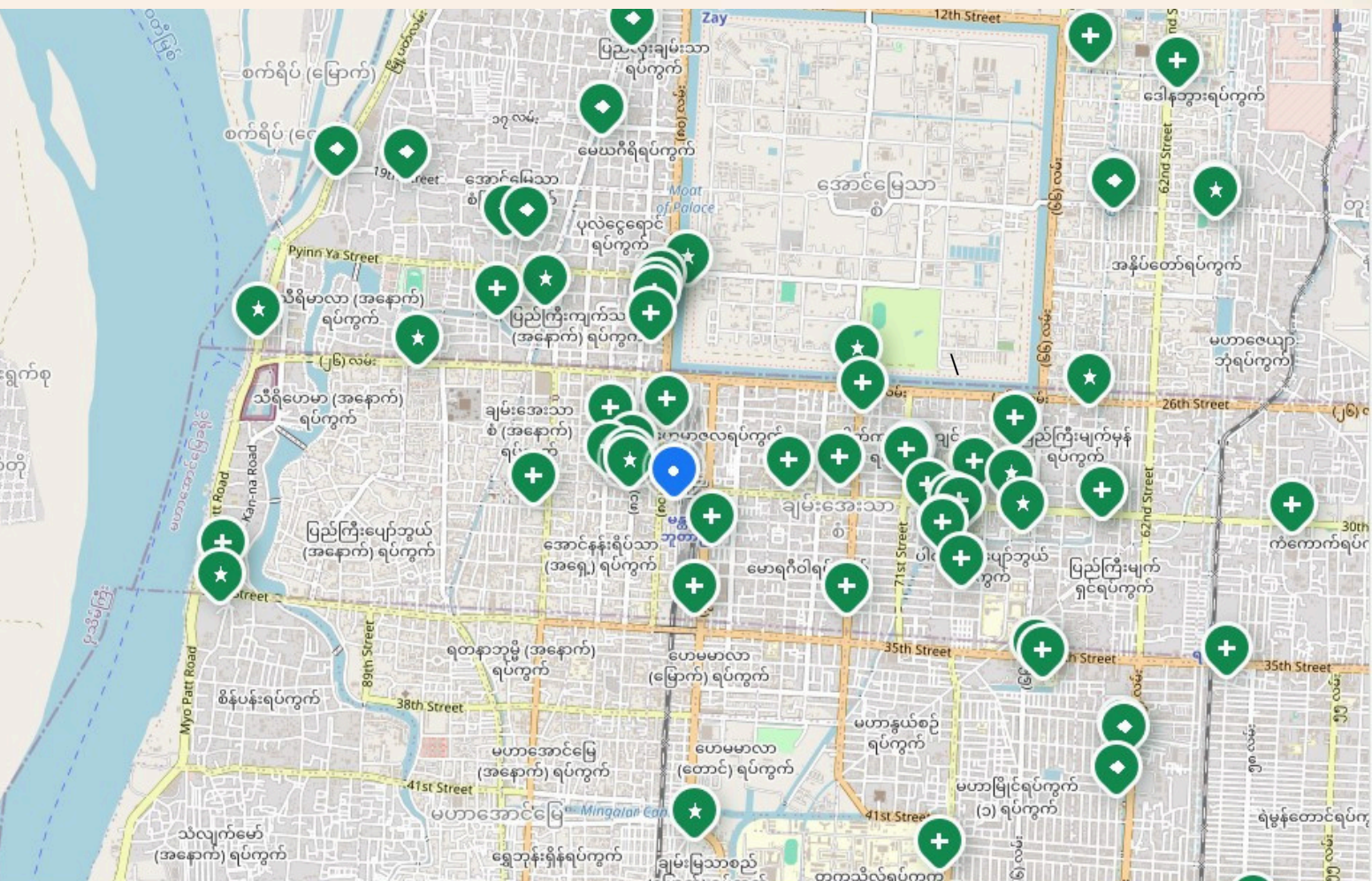
 **No 9**
police · 0.28 km

 **No.8 Myoma Police Station**
police · 0.29 km

 **Central Fire Station**
rescue · 0.32 km

12

System Recommend Nearest Support Center



Map showing the location of the user (blue dot) and the nearest recommended support center (green pin with plus sign). The map displays various streets and landmarks in Yangon, Myanmar.

Nearest locations

Location	Distance
No 9	0.28 km
No.8 Myoma Police Station	0.29 km
Central Fire Station	0.32 km

Police assistance

No 9
Police

DISTANCE
0.28 km

CONTACT
Not reported

[Get Directions](#)

[Compare & Recommend](#)

STATUS UNKNOWN

Navigation and Distance Estimation

The screenshot displays the OpenStreetMap website interface. At the top, the 'OpenStreetMap' logo is visible alongside navigation links: 'Edit', 'History', 'Export', 'GPS Traces', 'User Diaries', 'Communities', 'Copyright', 'Help', 'Donate', and 'About'. On the right, there are icons for a map and a user profile labeled 'saw bbb'.

The main map area shows a street grid with numerous red location pins. A blue line indicates a calculated route. The map includes labels for streets such as 28th Street, 29th Street, 30th Street, 31st Street, 32nd Street, 33rd Street, 34th Street, 84th Street, and 85th Street. Landmarks like 'Chan Hospital', 'Children's Clinic', 'Happy Cafe and noodles', and 'Royal Rose Hospital' are also labeled.

On the left side, the 'Directions' panel is active, showing a route with a distance of 659m and a time of 0:03. The steps are as follows:

1. Start on unnamed road
2. Continue on (၇၈) လမ်း
3. Turn left onto 32nd Street
4. Turn left onto unnamed road
5. Reach destination

Below the directions, there are links to 'Download route as GeoJSON', 'Directions courtesy of OSRM (FOSSGIS)', and 'More details from OSRM (FOSSGIS)'.

On the right side, the 'Map Layers' panel is open, showing a list of map styles: 'Standard', 'CyclOSM', 'Cycle Map' (which is highlighted with a blue border), 'Transport Map', 'Tracestrack Topo', 'Humanitarian', and 'Shortbread'.

At the bottom of the map, a scale bar indicates 200m and 500ft. The footer text reads: '© OpenStreetMap contributors. Tiles courtesy of Andy Allan. Website and API terms'.

Emergency Station Status Report

SafePath
REAL-TIME EMERGENCY RESPONSE

EN Admin

Admin mode: click the exact location for the new emergency camp.

All values are validated again by the protected admin API.

21.975256, 96.088990

Camp Name
SayTaNaMon

Population
10

Capacity
100

Food units
130

Medicine units
88

Status
Active ✓
Full
Closed

Contact
09236726748

Cancel Save Camp

SAFEPATH RECOMMENDATION
Help is within reach.
Review live location details, then open turn-by-turn directions.

Royal Rose Hospital
Hospital

DISTANCE
0.18 km

CONTACT
Not reported

Medical care

Get Directions

Compare & Recommend

Nearest locations
Royal Rose Hospital

Establishment of Emergency Rescue Station Notice

The screenshot displays the SafePath application interface. At the top left, the 'SafePath' logo is accompanied by the text 'REAL-TIME EMERGENCY RESPONSE'. On the top right, there are language and user controls: 'EN' and an 'Admin' button with a shield icon. The background is a map of an urban area with street names like '30th Street', '31st Street', '32nd Street', and '33rd Street'. Overlaid on the map is a white modal window. At the top of the modal, a dark green banner states 'Emergency camp saved and is now visible to users.' Below this, a text instruction reads 'Manage orange markers. Every saved camp is visible to all users.' The modal is divided into two main sections. The left section, titled 'SayTaNaMon' with a status of 'ACTIVE · 10/100 people', contains 'Edit' and 'Delete' buttons. The right section is titled 'Add a camp at an exact map location' and includes the instruction 'Click the button, then click once on the map where the orange marker should appear.' This section features a large green '+ Add Camp' button and a 'Sign out' link. To the right of the modal, a partial view of another screen shows a 'PATH RECOMMENDATION' section with the text 'is within reach.' and a 'STATUS UNKNOWN' label. Below this, 'Royal Rose Hospital' is listed with a 'CONTACT' button showing 'Not reported' and a 'Get Directions' button. At the bottom, a 'Nearest locations' section lists 'Royal Rose Hospital'.

Showing Emergency Rescue Station Location

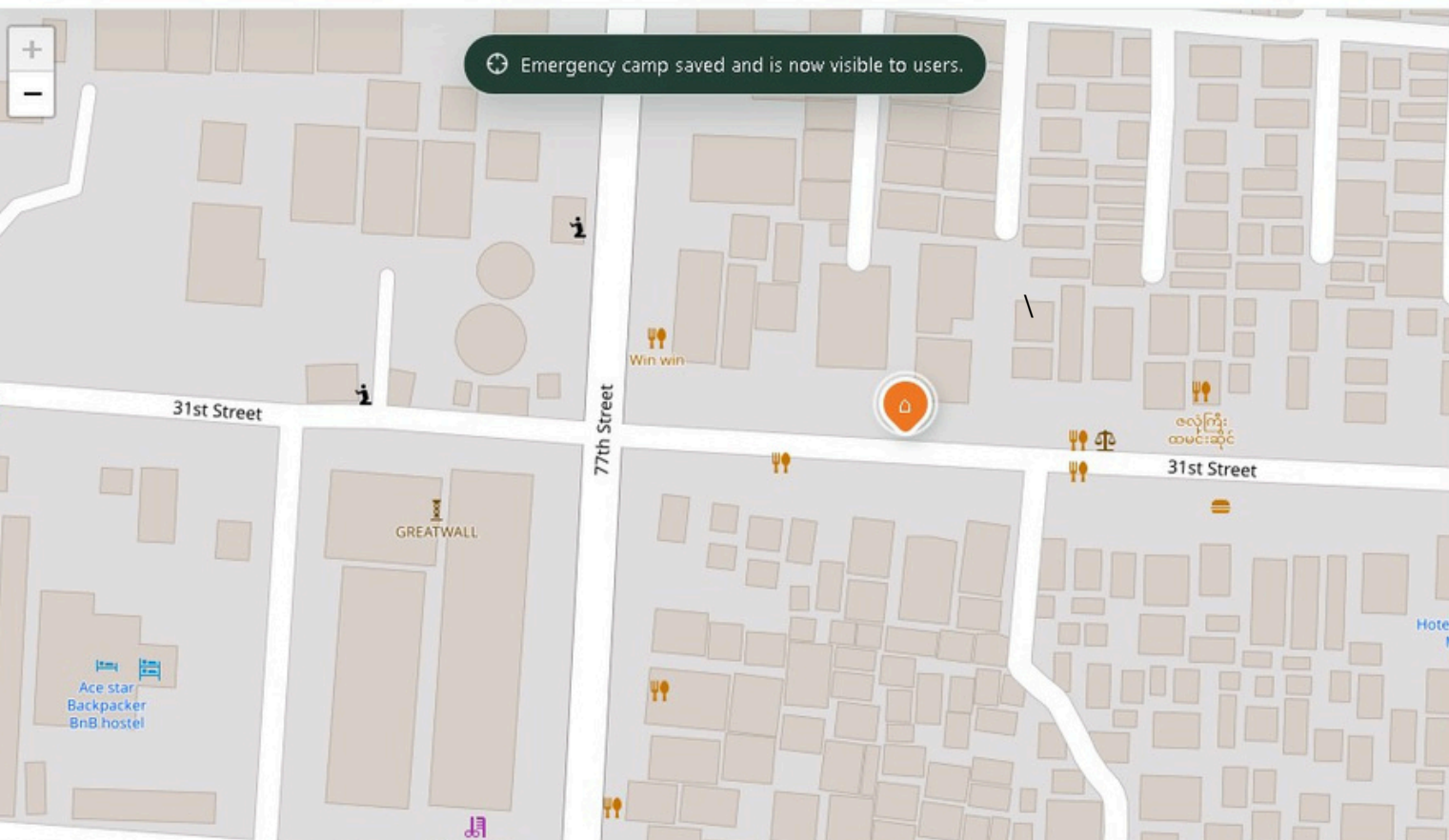
 **SafePath**
REAL-TIME EMERGENCY RESPONSE

EN Admin

+

-

Emergency camp saved and is now visible to users.



LOCATION DETAILS

Help is within reach.
Review live location details, then open turn-by-turn directions.

 **SayTaNarMon**
Emergency Camp

ACTIVE

DISTANCE
0.32 km

CONTACT
09236726748

POPULATION
10 / 100

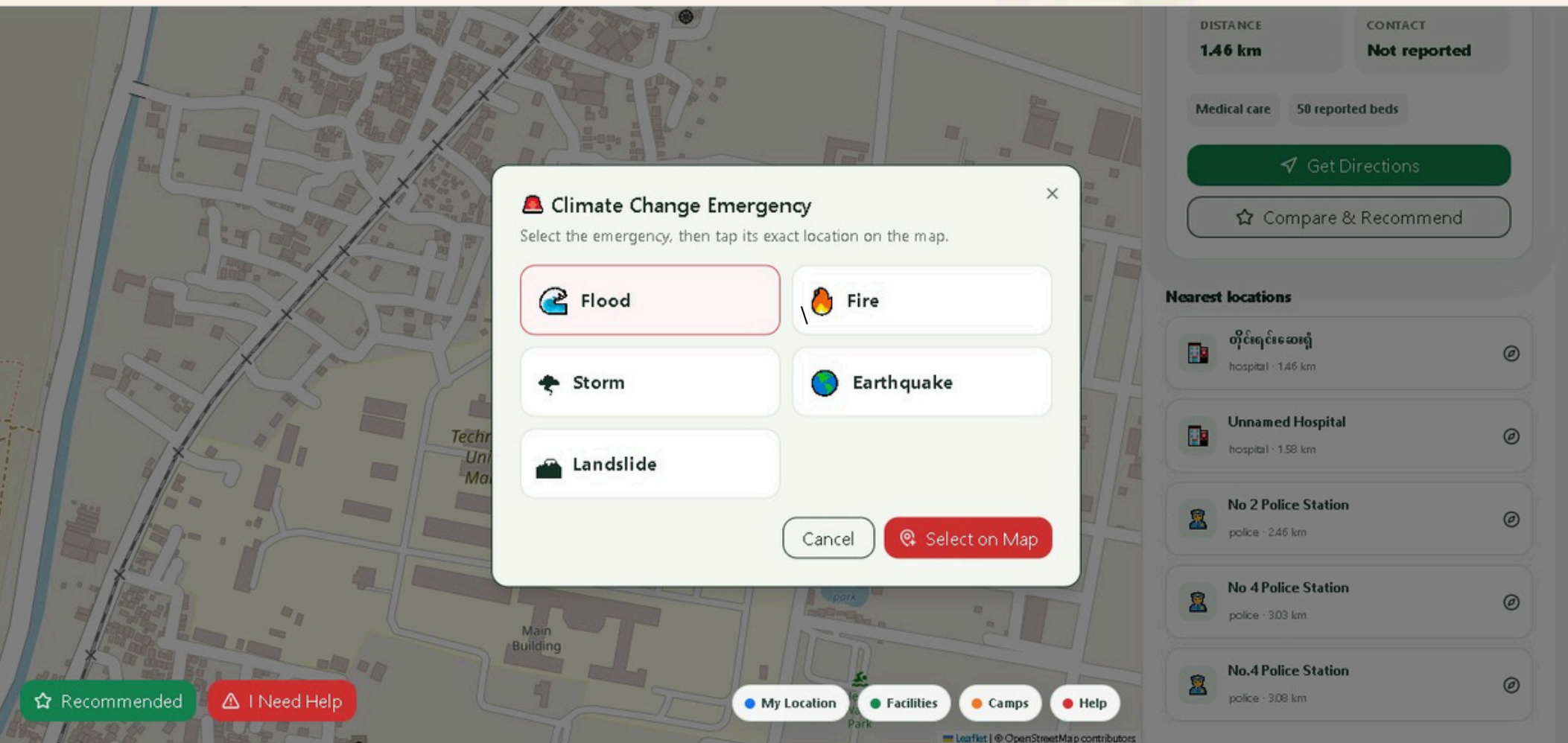
AVAILABLE SPACE
90

Food: 130 Medicine: 88 Space: 90

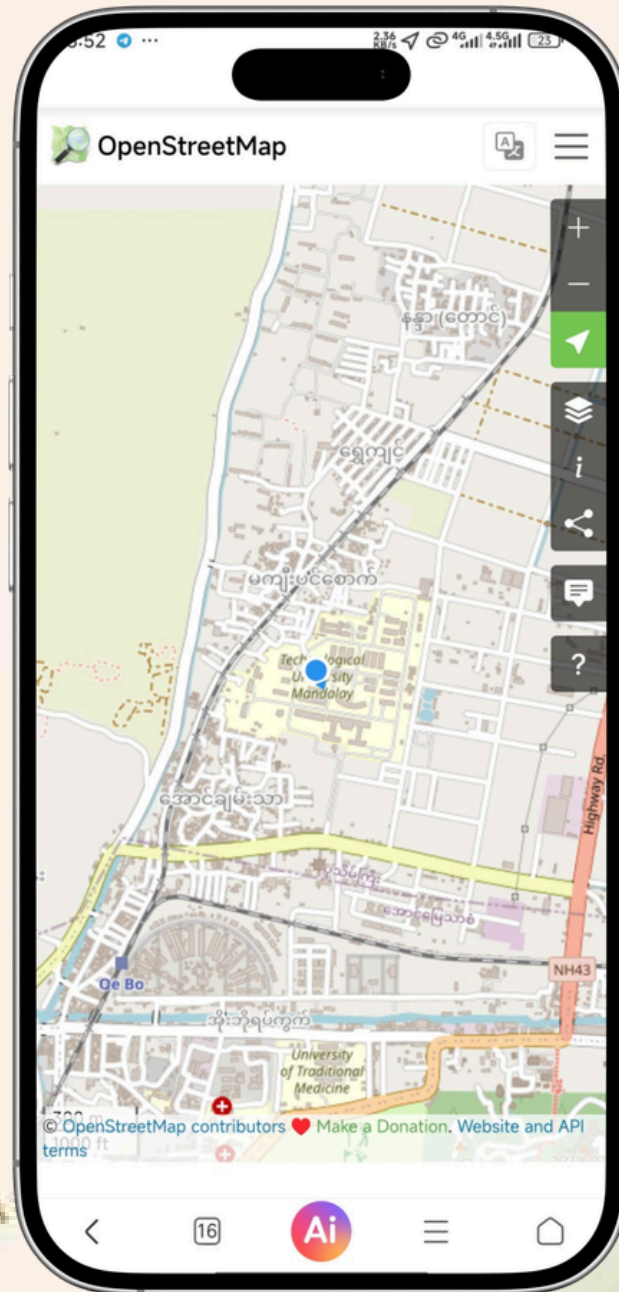
Get Directions

Compare & Recommend

Climate Emergency



Android Version



**When connection is off,
The last active location
stored and can direct
nearest Support Center.**

THANK FOR
YOUR
ATTENTION

TEAM ICONIX